Model re-entry — operations order (OPORD)

Last synced: 2026-07-27
Owner: Charles + Cursor agent (not a SCOUT handoff)
Status: Executed — archive, 540 surface, Pass B exports on disk.

Intent

Re-entry sim work mirrors the document re-entry pattern: **park old lab notebooks**, build a **thin** `540_*` **surface**, reuse `tier1_*` **engines**. Revolution = new problem approach and entry point; evolution = import battle-tested libraries.

Charles authored the three-step **pipeline**: assign → **S_i** → top **K**. Alex endorsed it.

User-facing assignment knob: ρ (rho) = assortativity (0 = max mixing; ρ↑ = sharper match). Legacy **τ (temperature)** in archived docs = inverse mixing; **ρ = 1, σ = 0.65** maps to old **τ ≈ 0.65**.

Phase 0 — Preconditions

#	Step	Owner
0.1	Proceed in this chat (Charles + Cursor agent)	Charles
0.2	Notebook burn: cell index 1 unless `skip burn`	Agent

Phase 1 — Spec on disk (brief before moves)

#	Step	Deliverable
1.1	Write `sports/540_READ_ME_SIM.md`	Re-entry sim contract
1.2	Pointer in `re_entry/00_READ_ME_FIRST.md`	One paragraph
1.3	Update `re_entry/02`, `03`: τ → **ρ** for assignment	Short patches
1.4	Update `PARKED_FOR_LATER.md`	538D archived; active = `540_*`

Gate: Charles approves `540_READ_ME_SIM.md` before Phase 2.

Phase 2 — Archive

#	Step	Deliverable
2.1	Create `sports/archive/` + `README.md`	Folder + manifest
2.2	Move into archive	`538D_development.ipynb`, `538_alex_tier1_model_and_fit.ipynb`, `537_Sports_Simulation.ipynb`, `535_sports_tier_1.ipynb`, `538D_widget_render_test.ipynb`, optional `535_*_outputs_backup_*.ipynb`
2.3	Grep and fix broken doc pointers	Docs we touch only
2.4	**Do not move**	`530`, `tier1_*.py`, `hero_model_reset_bundle.py`, `sports_pipeline/`

Phase 3 — COMPASS touch (light)

#	Step	Deliverable
3.1	`3-Master_Plan/20260727_COMPASS_sim_reentry_status.md`	½-page status + claims guard
3.2	Mirror plan if `~/.cursor/plans/` edited	`./scripts/mirror_plan.sh` (optional)

No mandatory COMPASS agent session.

Phase 4 — SCOUT (reference only)

#	Step	Deliverable
4.1	No SCOUT session required this sprint	—
4.2	Archived notebooks = SCOUT-era lab reference; do not edit	Note in `540_READ_ME_SIM.md`
4.3	Fresh SCOUT session only if `tier1_*` bug	Uses 540 brief

Phase 5 — Code (ρ + Pass B)

#	Step	Deliverable
5.1	ρ in `tier1_pool_assignment.py`	`exp(-ρ·(A_i-T_j)²/(2σ²))`; ρ=0 → uniform among open rosters
5.2	`USE_PREFERENTIAL_ATTACHMENT` bool in config	α=0 default
5.3	`sports/scripts/540_rho_ablation_bundle.py`	Low ρ, high ρ, sort-and-chop; fix selection
5.4	Export `alex_rho_ablation_v0/`	CSVs, PNG, summary, caption
5.5	`sports/540_three_step_sim.ipynb`	Thin Jupyter orchestration
5.6	Dry-run from repo root	Green run

Pass A (λ knockout): already in `alex_side_by_side_v0/` — do not redo.

Pass B (ρ ablation): fix **S_i** rule; vary assignment only.

Phase 6 — Reconcile Alex bundle

#	Step	Deliverable
6.1	Keep `alex_side_by_side_v0/`	Pass A
6.2	`alex_rho_ablation_v0/README.txt`	Cross-link to λ knockout
6.3	Limitation sentence for ρ pass	caption.txt

Phase 7 — Presentation to Alex (Charles leads)

#	Step	Deliverable
7.1	`Model.pdf` + 2 min addendum	Three-step pipeline; Pass A + Pass B
7.2	Minimal table	λ knockout vs ρ ablation arms
7.3	PDFs of updated docs	Charles runs convert scripts locally
7.4	Email / meeting	Figure paths + limitation sentences

Phase 8 — Close-out (when Charles asks)

#	Step
8.1	Git commit (docs + archive + scripts + tracked exports)
8.2	Park stretch in `PARKED_FOR_LATER.md`

Already done (skip)

- Empirical hero + LPM
- Pass A: λ knockout + `alex_side_by_side_v0/`
- Re-entry docs 01–03 + `Model.pdf`
- BINDING, unified **S_i**, **(B−D)=−L_C**

Execution order

```
1.1 → 1.4  →  2.1 → 2.4  →  3.1  →  5.1 → 5.6  →  6.1 → 6.3  →  7.x  →  8.x (optional)
```

Assignment formulas (540 contract)

Three-step assignment (generative step 1):

1. Draw **A_i** (abilities)
2. Draw **T_j** (team target means)
3. Soft place: $\pi_{ij} \propto \exp\left(-\rho\,(A_i - T_j)^2/(2\sigma^2)\right)$ × optional $(n_j + k)^\alpha$

ρ	Meaning
0	Uniform among teams with roster room (max mixing)
1	Moderate (≡ legacy τ=σ≈0.65)
High	Sharper match to **T_j** (not sort-and-chop)

Sort-and-chop: separate benchmark (`method="sort_chop"`) — max assortativity in 537 sense; not ρ→∞.

Selection (steps 2–3): S_i = A_i − λ·L_C; Pass B fixes λ, **L_C**, K.

Anti-patterns

- Adding cells to archived notebooks
- Duplicating `tier1_*` logic inside 540
- Claiming ρ ablation reproduces hero bin-for-bin
- Renaming ρ back to τ in user-facing prose

See also: `sports/540_READ_ME_SIM.md`, `3-Master_Plan/re_entry/00_READ_ME_FIRST.md`